
SRMIC-X1

64GB Residency-First Decode Architecture

Integrated Executive, Engineering, Packaging, and IP Specification

Version 1.0

Prepared for strategic technical discussion with hyperscale AI infrastructure stakeholders, including Meta-oriented deployment and architectural review.

This document is intentionally ambitious. It defines a flagship package target built around 64GB of distributed on-package SRAM, 64 logical residency regions, 4 compute chiplets, 8 HBM stacks, and a 96 TB/s class package fabric. It is written as a combined board paper, engineering architecture note, and patent-boundary draft.

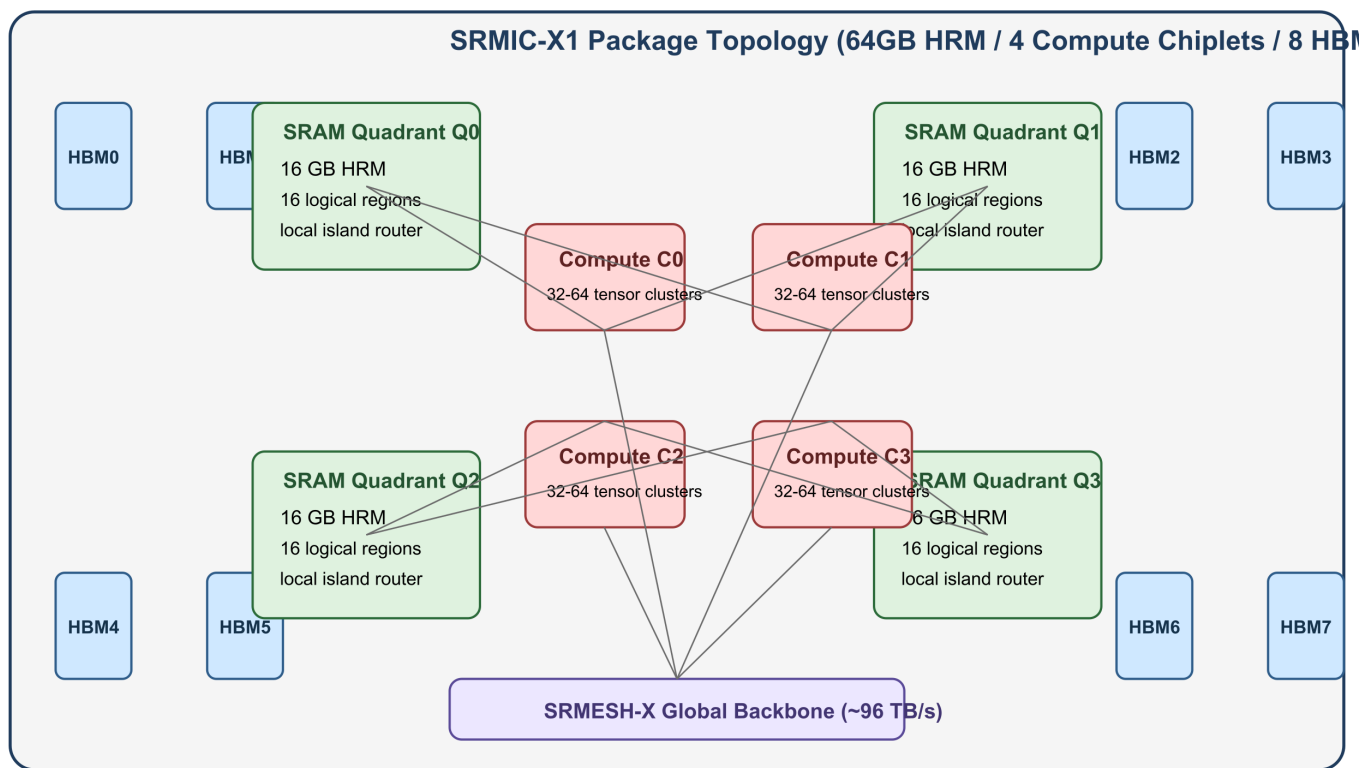
1. Executive Summary

SRMIC-X1 is a decode-optimized, residency-first inference accelerator built around a large distributed SRAM tier - Hot Residency Memory (HRM) - paired with HBM, an optional CXL warm tier, and a hierarchical package fabric named SRMESH-X. The architecture is designed to attack the dominant bottleneck in low-latency transformer decode: repeated movement of an active per-token working set from HBM.

The design target is not training and not broad GPU replacement. The intended use case is high-value, latency-sensitive transformer serving for 13B to 70B class models, plus staged larger-model deployments where the active decode working set is materially smaller than total model size.

Core thesis: if the active per-token working set resides in fast distributed SRAM and the package fabric bandwidth exceeds HBM bandwidth, decode latency collapses toward an SRAM-bounded floor rather than an HBM-bounded floor.

Flagship baseline	SRMIC-X1 target
HRM capacity	64 GB distributed SRAM
Logical regions	64 regions x 1 GB
Compute chiplets	4
HBM stacks	8
HBM bandwidth	~24 TB/s aggregate
SRMESH-X bandwidth	~96 TB/s target
Power envelope	1.0-1.5 kW class
Primary workloads	13B-70B decode, staged larger models



2. Why This Matters for Meta-Scale Serving

In Llama-scale deployment environments, decode often dominates end-to-end latency. Even in systems with substantial tensor compute, the per-token path repeatedly exercises memory movement for active weights, KV state, and bounded cold pulls. As model personalization, adapter use, and long-context sessions increase, the fraction of time spent waiting on memory grows rather than shrinks.

- HBM traffic dominates latency-sensitive decode more than tensor throughput.
- Batch sizes in interactive serving are often too small to fully amortize memory movement.
- Per-token latency predictability matters as much as aggregate tokens/sec.
- A large hot residency tier creates a structurally different performance regime than HBM-only decode.

3. Product Definition and Positioning

SRMIC-X1 should be positioned as a residency-first inference platform, not a general-purpose GPU and not a training-first architecture. It is explicitly designed for the decode half of transformer inference, where memory hierarchy, page residency, and bounded hot working sets matter more than universal programmability.

Dimension	SRMIC-X1 position
Primary objective	Reduce HBM bottleneck during decode
Best-fit workload	Low-latency transformer decode
Competitive wedge	Large SRAM residency tier + fabric > HBM
Software model	Residency-aware runtime / compiler hints
Not optimized for	General GPU graphics, broad training workloads

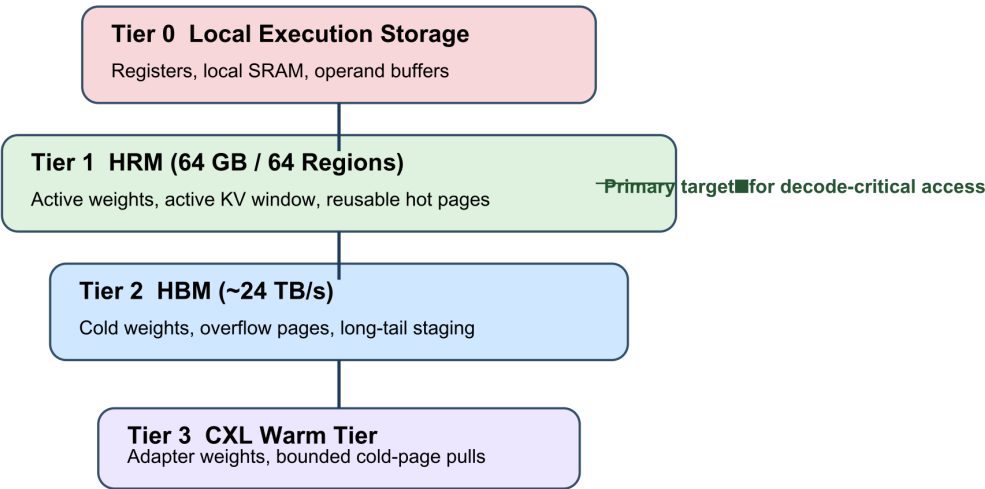
The clean strategic message for Meta is simple: SRMIC-X1 treats decode as a memory-system problem first and a tensor-compute problem second. That is the opposite ordering of most existing accelerators.

4. Competitive Framing

Competitor	Strength	SRMIC-X1 differentiation
NVIDIA-class GPU	Excellent HBM and training versatility	Decode-first hot-residency architecture, not training-first
Cerebras	Massive internal bandwidth / SRAM-chiplet thinking	Chiplet thinking modularity and package scalability
Groq	Deterministic scheduling and predictable latency	Residency and memory hierarchy are first-order design variables
Tenstorrent-class approaches	Distributed architectures and openness	Very large explicit SRAM hot tier sized for decode working sets

5. Top-Level System Architecture

SRMIC-X1 Memory Hierarchy and Decode-Critical Access Path



SRMIC-X1 defines a four-tier hierarchy. Tier 0 is local execution storage inside the compute subsystem. Tier 1 is HRM, the primary decode hot tier. Tier 2 is HBM, which stores cold and overflow pages. Tier 3 is a bounded CXL warm tier for adapters, long-tail state, or large-model assists.

- Tier 0: cluster-local registers and operand SRAM
- Tier 1: 64 GB distributed HRM, the primary target for decode-critical access
- Tier 2: HBM for cold weights and overflow active pages
- Tier 3: bounded CXL pulls for warm or adapter-like state

6. Package Baseline and Partitioning

The flagship package assumption is 64 GB minimum HRM implemented as macro-chiplets or stacked SRAM tiles, logically exposed as 64 regions. We recommend 8 SRAM macro-chiplets, each exporting 8 logical regions. The compute layer is formed by 4 compute chiplets. HBM is provided by 8 stacks distributed symmetrically around the package.

Component	Count	Logical role
Compute chiplet	4	Tensor clusters, schedulers, HBM controllers, fabric endpoints
SRAM macro-chiplet	8	8 logical HRM regions each
HBM stack	8	Cold / overflow / staging tier
CXL controller chiplet	0-1 optional	Warm tier ingress and bounded cold pulls

7. HRM Microarchitecture

Each HRM logical region is 1 GB of SRAM organized as a banked, independently serviceable residency island. The region owns a slice of the decode working set and participates in bounded page movement between HBM and CXL.

Region attribute	Target
Capacity	1 GB
Banking	Multi-bank, implementation-specific
Ports	At least 2 read-class ports (logical target)
Metadata	Residency tags, state bits, local stats
Local buffers	Promotion queue, bounded prefetch window
Endpoint	Direct region router ingress/egress

Architectural correction carried into the simulator: the per-region working set is bounded by model partitioning rather than by total occupancy. Higher residency raises hit ratio but does not increase the per-token amount of work each region must perform.

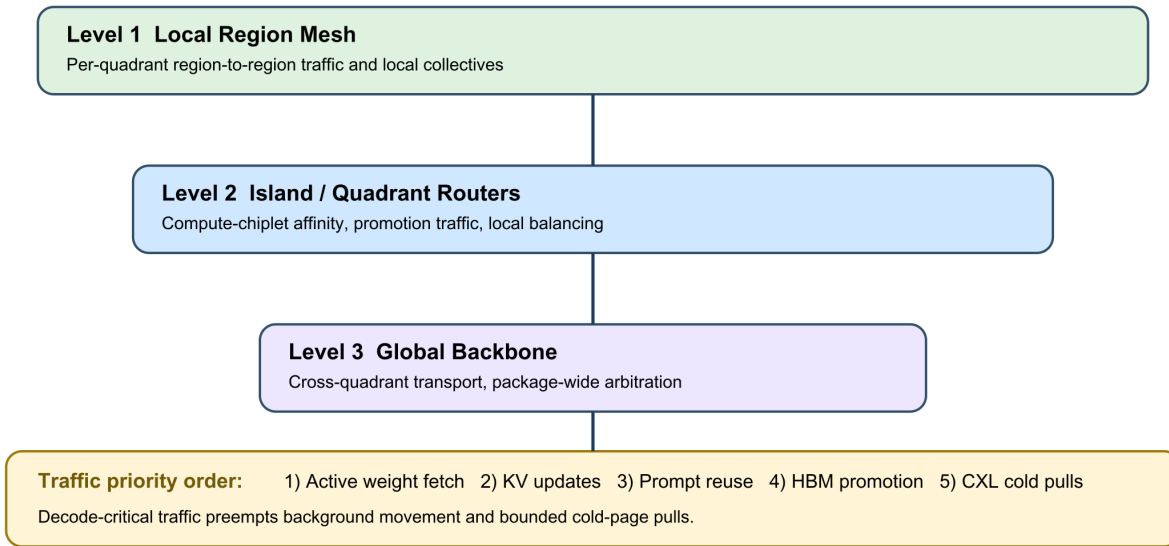
8. Compute Chiplet Microarchitecture

Each compute chiplet is a decode-specialized cluster complex. It includes 32-64 tensor clusters, a local scheduler, 2-4 lightweight control cores, fabric ingress/egress, HBM channels, and residency request logic. The package-level total is 128-256 tensor clusters.

- INT8 / FP8 decode-oriented execution
- Dedicated projection / GEMM paths
- Attention and KV update support
- Local scheduler aware of region placement and decode phase
- Control plane sufficient for orchestration, not oversized

9. SRMESH-X Fabric Architecture

SRMESH-X Hierarchical Fabric and Traffic Classes



SRMESH-X is explicitly hierarchical. A flat all-to-all package fabric becomes difficult to justify at 64 regions and 4 compute chiplets. The recommended organization is levelized.

Level	Role	Design intent
Level 1	Local region mesh	Region-to-region and local collectives inside an island
Level 2	Island / quadrant router	Compute affinity, local balancing, HBM promotion
Level 3	Global backbone	Cross-quadrant transport, package-wide arbitration

Traffic classes are prioritized with active weight fetch first, followed by KV updates, prompt reuse, HBM promotion, and finally bounded CXL cold pulls. This ordering is critical to maintaining decode-latency predictability.

10. Arbitration and QoS Model

At a minimum, each router should implement credit-based flow control, decode-priority queues, and weighted round-robin handling for non-critical traffic classes. The architecture benefits from explicit decode-phase tags so current-token requests are not forced to compete with maintenance traffic.

- Credit-based flow control to avoid buffer overruns
- Priority queues for decode-critical traffic
- Weighted RR for background promotion / spill traffic
- Optional multicast acceleration for shared prompt or adapter distribution
- Bounded arbitration delay target under normal operation

This document does not yet specify signal-level interfaces or exact arbitration cycle counts; it defines the service policy and the design requirement that decode-critical traffic must preempt background movement.

11. Decode-Step Timing Model

Single-Token Decode Timing Decomposition



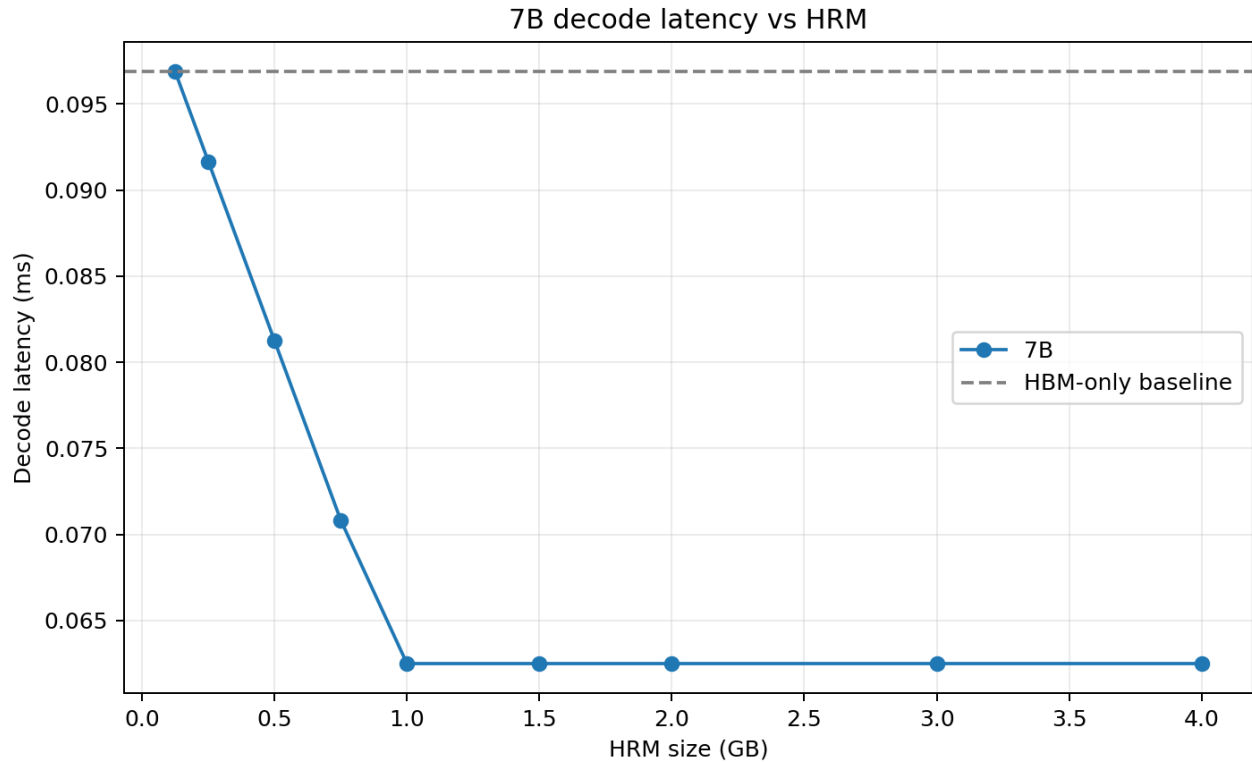
$$\text{Token latency} = \max(\text{compute}, \text{HRM}, \text{HBM}, \text{CXL}, \text{KV path})$$

Token latency is modeled as the maximum of compute time, HRM service time, HBM miss time, CXL page-window time, and the KV path. This correctly reflects the fact that the decode critical path is whichever service dominates the token step.

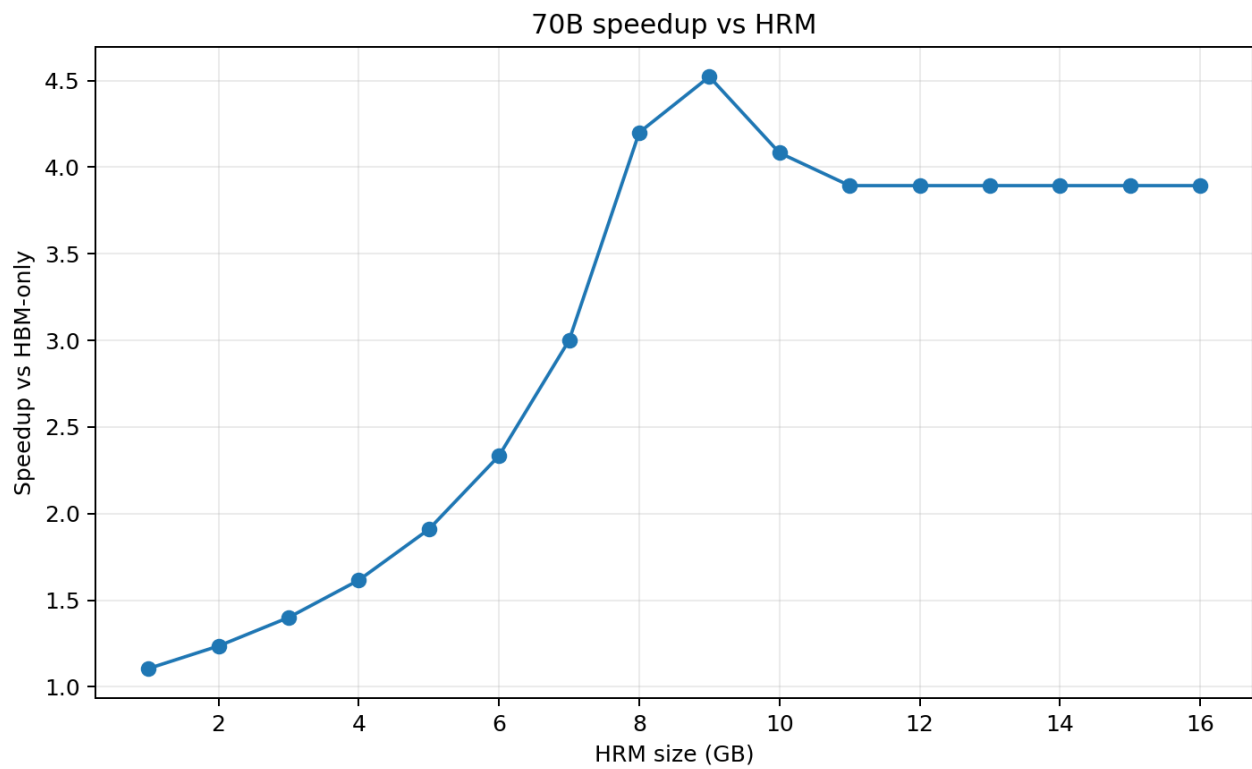
A key result from the simulation progression is that the model only becomes credible when HRM service is treated as parallel across fixed regional working sets rather than serialized through an aggregate pipe.

12. Simulation Findings

The architecture modeling journey matters because it defines what this design can honestly claim. Earlier versions of the model had artificial non-monotonic curves caused by incorrect serialization or occupancy-dependent HRM penalties. The current bounded working-set model removes those artifacts and yields the intended behavior: monotonic improvement with HRM until saturation, followed by flattening.

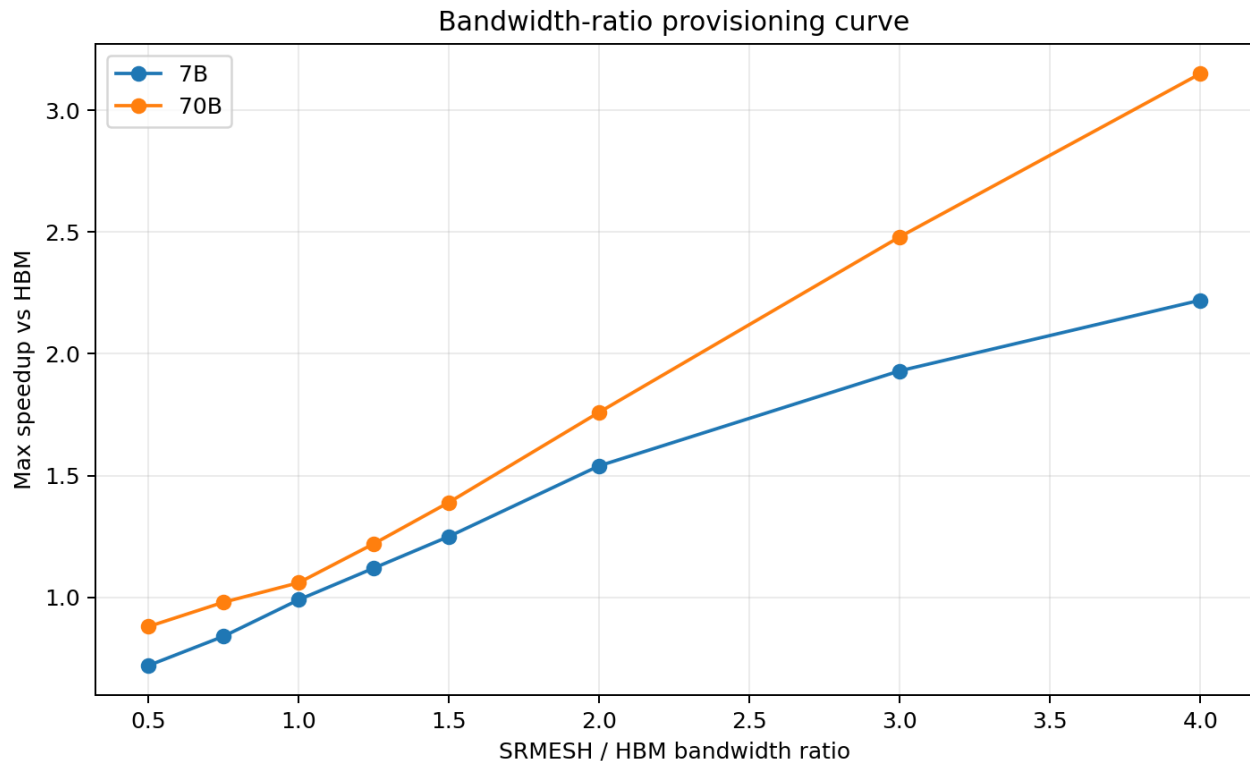


For compact models such as 7B under a deliberately constrained active fraction, HRM sensitivity is visible at sub-GB to low-GB scales. The latency curve improves as HRM approaches active-set coverage, then flattens.



For 70B-class models, speedup over HBM grows with HRM coverage until the active working set is effectively covered. There is no credible reason for the curve to degrade after that under the current architectural thesis; it should asymptotically approach the ratio imposed by the SRAM fabric and HBM bandwidth budget.

13. Bandwidth-Ratio Provisioning Curve

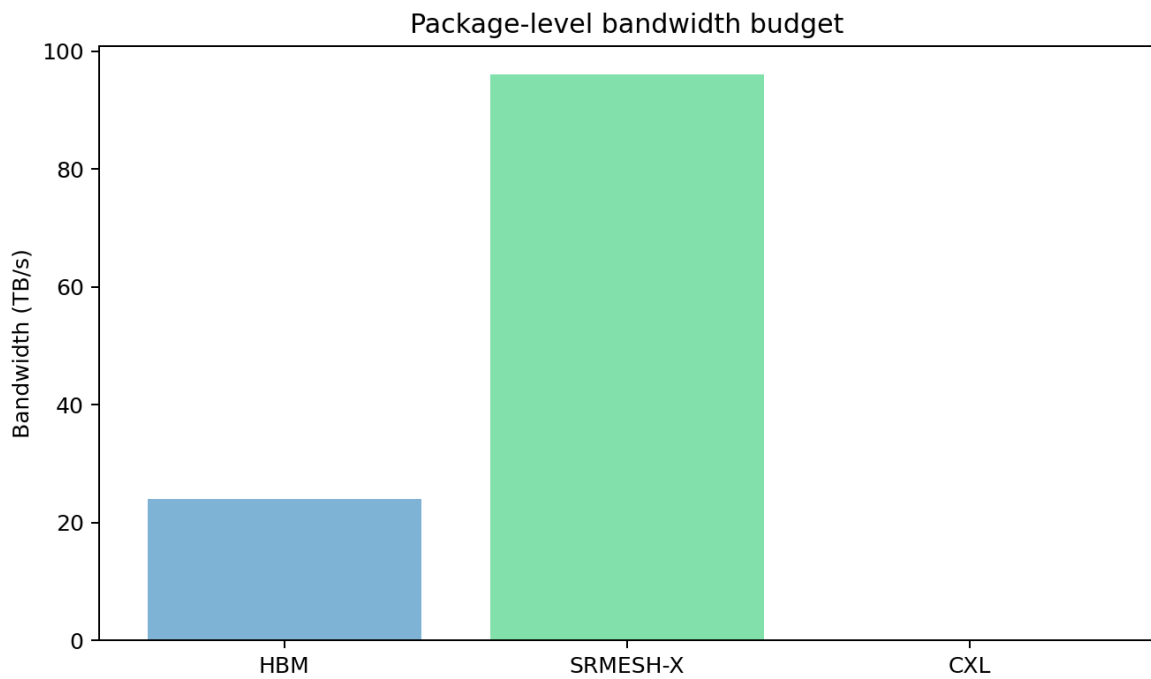


A central architectural result is that SRMIC-X1 only wins robustly when effective SRMESH bandwidth exceeds effective HBM bandwidth by a meaningful margin. The simulator suggests that the minimum viable ratio for broad competitiveness is above 1.0x and likely around 1.25x depending on workload, while stronger speedups require substantially more headroom.

Architectural invariant: SRMIC speedup requires SRMESH effective bandwidth > HBM effective bandwidth; this is not a tuning preference but a structural requirement.

The aggressive flagship target of 96 TB/s SRMESH against ~24 TB/s HBM aggregate produces ample room for saturation improvements and leaves headroom for arbitration, route inefficiency, and bounded CXL window costs.

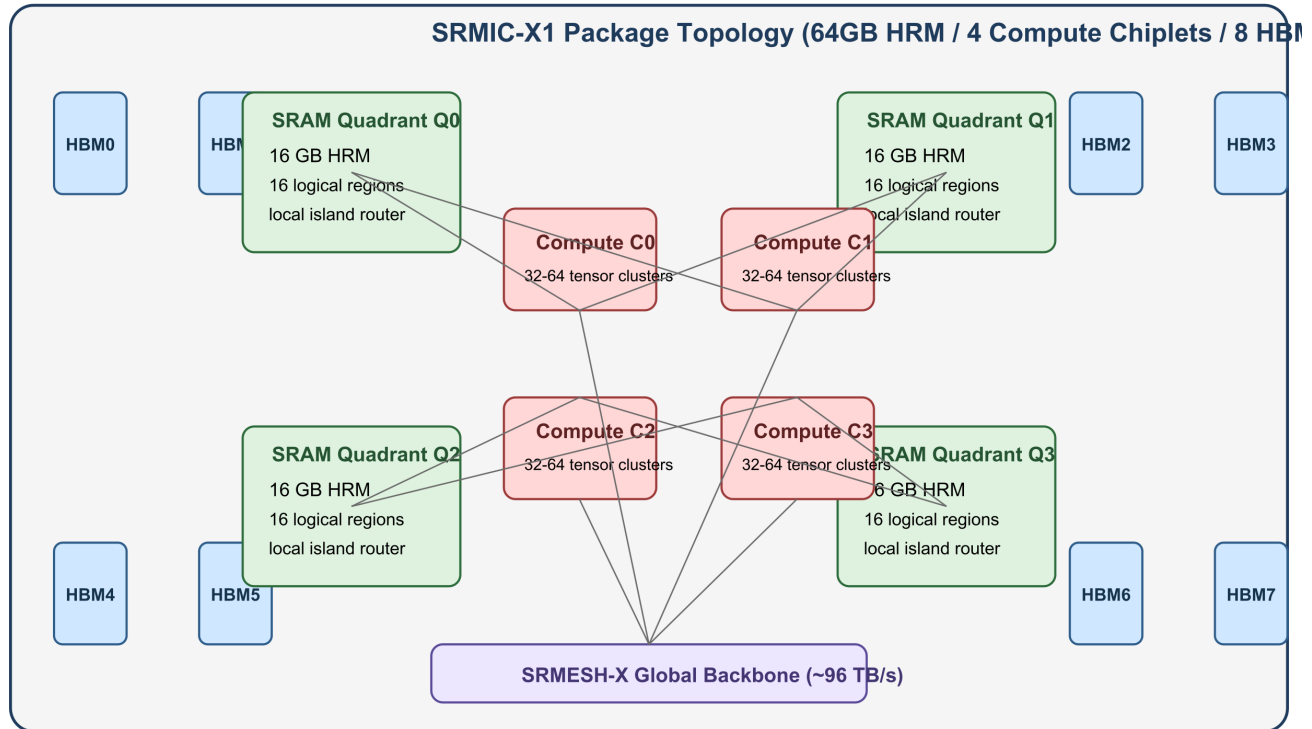
14. Package-Level Bandwidth Budget



At the package level, SRMESH-X is intentionally provisioned far above HBM. The design goal is not simply to match HBM but to make the hot residency tier unambiguously faster. If the hot tier is slower than the cold tier, the architecture loses at high residency and the thesis collapses.

15. Physical Package and Chiplet Concept

A 64 GB hot tier strongly implies a chiplet-oriented implementation. We recommend a package with 8 SRAM macro-chiplets, 4 compute chiplets, and 8 HBM stacks on an advanced interposer or comparable package substrate.



A natural physical organization is quadrant or octant, with each memory island combining local HRM regions, local router logic, and HBM affinity. The compute chiplets are positioned to maintain strong affinity to multiple islands while keeping average hop count bounded.

Physical block	Recommended role
SRAM macro-chiplet	Expose 8 logical 1 GB regions with local router
Compute chiplet	32-64 tensor clusters, control plane, HBM channels, route endpoints
HBM stack	Cold and overflow tier with adjacency to memory islands
Backbone fabric	Cross-quadrant transport and package-level arbitration

16. Thermal and Power Zoning

At 1.0-1.5 kW package power, floorplan and thermal zoning are first-order concerns. Compute chiplets should be placed centrally or symmetrically to distribute heat, while SRAM macro-chiplets should be organized to avoid hot-spot stacking. HBM stacks need balanced placement for power delivery and signal integrity.

- Central or symmetric compute placement for heat spreading
- Symmetric HBM distribution for package balance
- SRAM quadrants arranged to minimize average route length

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- Independent voltage and power-monitor domains for compute vs HRM
 - Thermal telemetry hooks required from day one

17. Area and Feasibility Considerations

A full 64 GB SRAM tier is expensive in area, packaging complexity, and leakage budget. This document does not pretend otherwise. The feasibility story depends on modular SRAM macro-chiplets or stacked SRAM tiles rather than a monolithic die. That is a feature of the architecture, not a weakness.

Feasibility question	Current answer
Can 64 GB fit monolithically?	No credible assumption suggests this is the right path
Can chiplets carry it?	Yes, with aggressive packaging and macro partitioning
Does the model require 64 GB exactly?	No, but flagship positioning benefits from this capacity
Can the architecture scale down?	Yes, 16-32 GB validation variants are plausible

A useful practical roadmap is to validate the design philosophy on a smaller package while keeping SRMIC-X1 as the flagship north star. That preserves ambition while creating a believable intermediate step for implementation risk management.

18. Software and Runtime Contract

SRMIC-X1 requires software cooperation. A runtime or compiler layer must characterize the active working set, map it to regions, manage bounded page-window pulls, and maintain promotion/demotion policy. This can begin software-heavy and become more hardware-assisted in later revisions.

- Layer and phase-aware residency hints
- Bounded page-window prefetch guidance
- Region mapping for active weight partitions
- Demotion and spill thresholds
- Telemetry export for hit ratio and route pressure

19. IP / Novelty Boundaries

Several potential invention boundaries are already visible from the modeling and architecture work.

- Bounded per-region working-set model for transformer decode residency
- Hierarchical fabric with decode-priority arbitration across HRM/HBM/CXL tiers
- Large-scale distributed SRAM tier explicitly sized for per-token active working-set service
- Residency-aware decode scheduling with promotion/demotion assists
- Hardware contract that treats CXL as a bounded page-window tier rather than a streaming cold-model tier

These boundaries are strongest when written as system claims, fabric claims, and runtime-cooperation claims together, rather than as isolated memory claims.

20. Risk Register

Risk	Why it matters	Mitigation direction
SRAM cost and area	64 GB is packaging-heavy and expensive	Macro-chiplets, staged validation path
Fabric congestion	Decode-critical traffic can be delayed by main memory traffic classes	Priority traffic classes, bounded background movement
Yield and assembly complexity	Large package with many active dice	Modular redundancy, phased bring-up plan
Software integration	Residency strategy requires runtime support	Compiler/runtime co-design from outset
Competitive response	Incumbents may add more SRAM or better DRAM	Differentiating on architecture invariant and package-level memory system

21. Why This Is Meta-Relevant

For a Meta-facing discussion, the strongest message is not that SRMIC-X1 is a universal accelerator. The strongest message is that it is purpose-built for the decode bottlenecks that matter in serving large open-weight models at scale. The architecture is well-aligned with Llama-family inference, adapter-heavy personalization, and long-context interactive serving.

- Direct attack on decode latency rather than broad GPU generality
- Memory-system-first architecture for serving economics
- Potentially better energy per generated token for memory-bound decode
- Clear route to model-specific optimization and residency-aware runtime control

22. Recommended Meta Pitch Framing

The external pitch should emphasize three ideas:

- SRMIC-X1 converts decode from an HBM problem into an SRAM residency problem.
- The architecture only wins when the fabric is provisioned above HBM - and we designed it that way.
- The package is intentionally flagship-grade because the opportunity is in large-model serving, not small convenience accelerators.

A weak pitch says: 'here is a chip with lots of SRAM.' A strong pitch says: 'we identified the decode bottleneck, derived the invariants required to beat HBM, and designed the package around those invariants.'

23. Implementation Roadmap

Phase	Goal
A	Lock architecture, invariants, and package baseline
B	Build area/power feasibility model and chiplet floorplan
C	Define fabric interfaces, route classes, and region protocol
D	Extend simulator with contention, bank conflicts, and longer-context stress
E	Produce investor / board / partner materials and patent package

A sensible near-term path is to preserve SRMIC-X1 as the flagship architecture while also defining a smaller validation derivative. The flagship keeps the strategic story compelling; the derivative keeps the implementation plan credible.

24. What This Document Does and Does Not Claim

This document claims that the residency-first architecture is structurally compelling, that the bandwidth invariant is real, and that a 64 GB HRM package is a coherent flagship target. It does not yet claim that all packaging, yield, thermal, or software challenges have been solved. It is intentionally ambitious, but it is not pretending that ambition removes engineering risk.

25. Closing Statement

SRMIC-X1 is the strongest form of the idea developed through the simulator work: a 64 GB, 64-region, chiplet-based, residency-first decode machine whose performance is governed by a clear architectural law - effective hot-tier bandwidth must exceed effective cold-tier bandwidth by a meaningful factor. If that law is satisfied and the active working set is bounded, the architecture has a path to materially lower decode latency than HBM-dominant accelerators.

That is the core of the pitch, the engineering thesis, and the patent story.

Appendix A. Formal Invariants

Invariant	Description
I1	Effective SRMESH bandwidth must exceed effective HBM bandwidth.
I2	Per-region working set is bounded by model partitioning, not total residency size.
I3	Decode-critical traffic must preempt background movement.
I4	CXL is modeled as bounded page-window traffic, not full cold-model sweep.
I5	Peak performance saturates once HRM covers the active working set.

These invariants are the compact form of the architecture. If any one fails, the performance thesis weakens materially.

Appendix B. Example 70B Working-Set Budget

Parameter	Illustrative value
Model size	70 GB
Active fraction	15%
Active working set	10.5 GB
HRM regions	64
Working set per region	164 MB
SRMESH-X aggregate target	96 TB/s
Per-region effective service target	1.5 TB/s
HBM aggregate	24 TB/s

This appendix is intentionally first-order. It exists to make the order of magnitude legible and to ground the claim that the per-region working set is manageable if the package is provisioned correctly.

Appendix C. Decode Priority Traffic Contract

- Current-token active weight requests must have highest priority.
- KV updates rank next because they immediately affect subsequent token steps.
- Prompt reuse and bounded multicast traffic may be admitted behind decode-critical paths.
- HBM promotion and CXL pulls should be backpressured aggressively if decode-critical queues rise.

This traffic contract is what translates the abstract bandwidth target into a latency target.

Appendix D. Open Engineering Questions

- Optimal bank organization per 1 GB HRM region

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- Exact route width and clock targets for 96 TB/s aggregate fabric
 - Trade-off between 4 and 8 compute chiplets
 - Whether 8 SRAM macro-chiplets or finer partitioning is more manufacturable
 - How much hardware assistance is needed versus software-managed residency
 - How to integrate low-overhead telemetry for hit ratio and congestion reporting